

DAVID KURNIAWAN

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EDUCATION

Xiamen University Malaysia | Selangor, Malaysia

Expected Sept 2028

BEng (Hons) in Artificial Intelligence (GPA: 3.84 out of 4.00)

Dean's List Awardee for 3 consecutive semesters — College of Artificial Intelligence and Robotics • Top 16% of cohort

TECHNICAL SKILLS

Languages: Python • HTML & CSS • JavaScript • C Programming • C++

ML / AI Frameworks: PyTorch • TensorFlow • Scikit-learn • Pandas • LangChain • OpenCV

Infra and Frameworks: FastAPI • Flask • Node.js • React • PostgreSQL (pgvector) • Docker • GitHub Actions • Git

Languages (Human): Indonesian (Native) • English (IELTS Band 7) • Chinese (Elementary)

SELECTED PROJECTS

[KerjaCerdas](#) | LangGraph, FastAPI, PostgreSQL (pgvector), Docker, React

2026

Lead Engineer / Backend

- Architected an **AI job-matching platform for Indonesian job seekers and small businesses** that ranks candidates by proven skills rather than CV keywords, built on **FastAPI** with a **LangGraph** response layer and React 18.
- Engineered an explainable, evidence-weighted ranking engine (semantic similarity 35%, skills 40%, experience 15%, education 10%) over **768-dim Gemini embeddings** and **pgvector (HNSW)**, where each skill is weighted by proof level: CV claim, passed skill quiz, or employer-confirmed after interview.
- Built the **seeker journey (UX)** from target job to skill gap, learning, application, and rejection feedback, plus a bulk job-posting uploader that parses PDF job packs into structured listings with Gemini.
- Implemented privacy and cost safeguards: **regex-based PII redaction** before any LLM call, consent prompts for scanned CVs, a 3-model fallback chain, and a token-efficiency gate that skips low-value LLM calls.
- Hardened a 5-gate **GitHub Actions CI pipeline** with gating integration tests against a live pgvector database and Alembic migration round-trip checks.

[Orion](#) | FastAPI, LangGraph, LangSmith, Pydantic, GitHub Actions

2026

Tech Lead / Backend (4-person team)

- Built a 6-stage autonomous workflow with **LangGraph** and **FastAPI** that parses, validates, and auto-approves enterprise expense claims, reducing processing time by ~80%.
- Implemented a deterministic policy engine in **Python with rapidfuzz and pypdf** for fuzzy-match duplicate detection and audit logging, with **Pydantic** type contracts between stages.
- Integrated **LangSmith** for tracing, token cost monitoring, and debugging of agent execution across all stages. Established **CI/CD in GitHub Actions**: PR gates enforcing 85% code coverage, plus nightly regression suites running against live LLM APIs.

[Indonesian Legal RAG](#) | LangChain, FAISS, BM25, Hugging Face, Weights & Biases

2026

- Built a hybrid retrieval pipeline (**BM25 + FAISS dense search, HyDE** query expansion, **cross-encoder** reranking) that returns exact statutory article citations for Indonesian labor law, with a live web search fallback when reranker confidence is low.
- Fine-tuned Llama-3-8B on a legal dataset**, tracked experiments with Weights & Biases, and published the open-weights model to Hugging Face.

AWARDS AND CERTIFICATIONS

Silver Award — SEA-CICSIC 2026

June 2026

China-ASEAN Innovation Competition (Undergraduate Category)

- Led AI strategy and technical architecture for Omni-QC, a manufacturing platform for real-time defect detection and predictive quality control.

3rd Place — DPickleball AI Competitions (Unity • ML-Agents • Reinforcement Learning)

Oct 2025

XMUM AI Club • Xiamen University Malaysia

- Lead developer of a 3-member team; designed the reward shaping and PPO training pipeline using Unity ML-Agents.

Top 20% Globally — International Quant Championship, Stage 1

Apr 2025

World Quant BRAIN